

1. Run `task_0a.py` to generate the vector database if not already generated.
2. Press Run All (or restart kernel and run all cells).
3. You will be prompted to provide input values.

```
In [ ]: LABEL = input("Provide a valid label name / label ID present in the Caltech101 dataset\n")  
  
        FEATURE_SPACE = input("Provide a feature space [color, hog, avgpool, layer3, layer4, layer5]\n")  
  
        K = int(input("Enter K, the number of similar images K to find for the selected label\n"))
```

```
Input label: 0
Downloading dataset if not present.
Files already downloaded and verified
Output label: Faces
```

```
In [ ]: # Load feature space and representative label vectors for that feature space

from utils.database_utils import retrieve

feature_vectors = retrieve(f'{FEATURE_SPACE}.pt')
rep_label_vectors = retrieve(f'rep_label_{FEATURE_SPACE}.pt')
```

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```
for i in range(len(top_k)):
    distance, image_id, label = top_k[i]

    print(str(i + 1) + ".", "\tImage ID: ", image_id, "\t\tDistance: ", distance)
    display(dataset[image_id][0])
```

Downloading dataset if not present.
Files already downloaded and verified

K = 10 most similar images by hog feature space using distance: correlation

1. Image ID: 7684

Distance: 0.02843330077030426



2. Image ID: 7354

Distance: 0.03296948516724818



3. Image ID: 7696

Distance: 0.03483804447323913



4. Image ID: 6922

Distance: 0.03684505768056978



5. Image ID: 6420

Distance: 0.03696405885847864



6. Image ID: 5390

Distance: 0.03727268476412071



7. Image ID: 7000

Distance: 0.03819314140726948



8. Image ID: 104

Distance: 0.03982271060610387



9. Image ID: 3402

Distance: 0.042002216887863764



10. Image ID: 5574

Distance: 0.042174124807494495



